A Nonlinear Two-Path Regression Test in Factor Asset Pricing Models

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1 Empirical Studies

Let Y_{it} be the excess returns, F_t be the k-dim factors. We assume the market model:

$$Y_{it} = a_i + b_i' F_t + \varepsilon_{it} \tag{1}$$

Fiorentini, Galesi, and Sentana, 2018 Many economic theories imply that the expected return is a linear function of betas:

$$E(Y_i \mid b_i) = b_i' \gamma \tag{2}$$

and we test this relationship.

We have two sets of data:

1. Factor zoo data and monthly return data for all CRSP stocks
   (a) Factor zoo data from Feng, Giglio, and Xiu, 2017, 1976-06 to 2017-12, monthly data of 147 factors.
   (b) Monthly holding-period return data of 463 stocks on CRSP that don't have NA in the sample period. The total number of available stocks are 29,742, but (too) many have NA.
2. CRSP Beta Suite dataset:

   For SP500 constituents from 1990-01-02 to 2019-12-31, we have daily data on
   (a) return;
   (b) betas on Carhart's four factors, calculated using moving average, so the betas have variation.
   (c) idiosyncratic and total volatility.

   The daily risk-free rate is taken from French's website. After dropping NA, we are left with 253 stocks, but it can be expanded to incorporate more stocks.

1.1 Test Procedure

1. For each $i = 1, \ldots, n$, I did the following regression, let Y_{it} be the excess return $R_{it} - R_t^f$, let F_t be k-dim factor excess return

$$Y_{it} = a_i + b_i' F_t + \varepsilon_{it} \tag{3}$$

   obtain estimates $\hat{a}_i, \hat{b}_i$;

2. The cross-sectional regression of $\bar{Y}_i = \frac{1}{T}\sum_t Y_{it}$ on $\hat{b}_i$. If we include the intercept, then we are only testing that the expected return is a linear function of betas, i.e., there exist $(\alpha, \gamma) \in \mathbb{R} \times \mathbb{R}^k$ such that

$$E(Y_i \mid b_i) = \alpha + b_i' \gamma \tag{4}$$

whereas if we do not include the intercept in this step, we are testing for both linearity and no significant pricing error, there exists γ:

$$E(Y_i \mid b_i) = b_i'\gamma \tag{5}$$

In either case, we obtain the estimate $\hat{\gamma}$ and the residual $\hat{u}_i = \bar{Y}_i - \hat{b}_i'\hat{\gamma}$.

3. Collect $(\hat{u}_i, i = 1, \ldots, n)$ and construct a specification test statistic as in Li and Racine, 2007, which is very close to Hardle and Mammen, 1993. First we calculate

$$I_n := \frac{1}{n(n-1)} \sum \hat{u}_i \hat{u}_j K_{h,ij} \tag{6}$$

where $\hat{K}_{h,ij} = \prod_{s=1}^{k} h_s^{-1} K\left(\frac{\hat{b}_{is} - \hat{b}_{js}}{h_s}\right)$, I used Guassian density $K(\cdot)$ and the bandwidth for the s-th beta is chosen to be $h_s = 1.06 \times \text{std}(\hat{b}_s) \times n^{-1/5}$ for simplicity. I_n corresponds to the hypothesis:

$$H_0 : E(uE(u \mid b)f(b)) = 0, a.s. - b \tag{7}$$

The test statistic is

$$T_n = \frac{n(h_1 \ldots h_k)^{\frac{1}{2}}}{\sqrt{\hat{\sigma}^2}} I_n \tag{8}$$

where $\hat{\sigma}^2 = \frac{2(h_1 \ldots h_k)}{n(n-1)} \sum_{i \neq j}^{n} \hat{u}_i^2 \hat{u}_j^2 \hat{K}_{h,ij}^2$

4. Use wild bootstrap to approximate the distribution of T_n.

1.2 Factor Zoo Individual Test

With the first set of data, we did the following tests, with Factor X taking each factor in the zoo:

1. Test each factor individually, that is we assume there is only one Factor X.
2. Test Market Factor + Factor X.
3. Test FF3 Factors + Factor X.

In the second-step cross sectional regression, we have incorporated an intercept, so the test is purely for nonlinearity.

	Factor X	MktRf + Factor X	FF3 + Factor X
Number of Rejections	34	30	41

Table 1: Number Of Rejections for Different Models

Question 1. I think the reason model $FF3 + X$ has more rejections is because if we use full sample, then MktRf and HML are both rejected for being linear. I wonder if it makes more sense to split the sample.

The following is the table of result. The numbers are 1. test statistic value and bootstrap distribution quantiles. Red ones are those rejected.

Name of Factor X	Factor X	MktRf + Factor X	FF3 + Factor X
	(T_n, Bootstrap 2.5% and 97.5% quantile)		
BAB	(0.687,-1.443,1.636)	(2.975,-1.622,1.803)	(2.820,-1.778,2.216)
CMA	(-0.858,-1.400,1.083)	(0.516,-1.549,1.532)	(0.274,-1.797,1.777)
HML	(2.126,-1.453,1.458)	(0.014,-1.742,1.579)	(—,—,—)
HML_Devil	(2.230,-1.461,1.918)	(0.812,-1.644,1.613)	(2.006,-1.884,1.709)
HXZ_IA	(-0.134,-1.429,1.133)	(0.498,-1.556,1.317)	(0.767,-1.922,1.266)
HXZ_ROE	(0.633,-1.389,1.552)	(2.659,-1.641,1.539)	(1.732,-1.835,2.028)
IPO	(-0.757,-1.463,1.224)	(2.338,-1.767,1.651)	(2.450,-1.871,2.175)
LIQ_PS	(3.033,-1.385,1.360)	(1.878,-1.635,2.289)	(0.725,-1.729,2.027)
LTR	(-0.952,-1.463,1.099)	(1.522,-1.698,1.550)	(1.310,-1.865,1.561)
MktRf	(2.144,-1.450,1.415)	(—,—,—)	(—,—,—)
QMJ	(-0.778,-1.478,1.358)	(1.513,-1.736,2.028)	(1.095,-1.760,1.789)
RF	(2.462,-1.420,0.964)	(1.671,-1.760,2.044)	(1.376,-1.733,1.657)
RMW	(0.203,-1.421,1.132)	(0.397,-1.737,1.420)	(2.693,-1.689,1.705)
SMB	(-0.185,-1.494,1.267)	(0.510,-1.768,1.706)	(—,—,—)
STR	(1.219,-1.480,1.274)	(3.128,-1.830,1.456)	(0.520,-1.906,1.571)
UMD	(0.332,-1.384,1.131)	(0.018,-1.706,1.611)	(1.071,-1.843,1.997)
absacc	(1.299,-1.324,1.460)	(0.956,-1.617,2.012)	(2.351,-1.847,1.576)
acc	(7.448,-1.469,2.029)	(2.096,-1.720,1.597)	(1.682,-1.928,1.941)
adm	(1.192,-1.237,1.564)	(1.689,-1.514,1.599)	(1.229,-1.886,1.805)
aeavol	(-0.755,-1.326,1.199)	(-0.238,-1.502,1.712)	(0.127,-1.795,2.034)
age	(-0.684,-1.379,1.255)	(0.494,-1.654,1.638)	(2.300,-1.860,1.451)
ala	(-0.330,-1.482,1.741)	(0.737,-1.646,1.712)	(1.523,-1.680,2.094)
ato	(-0.837,-1.344,1.442)	(2.233,-1.774,1.864)	(1.945,-1.870,1.560)
baspread	(0.475,-1.347,1.156)	(1.510,-1.618,1.806)	(3.327,-1.776,1.746)
beta	(0.554,-1.375,1.392)	(2.441,-1.608,1.810)	(3.255,-1.655,1.787)
bm_ia	(0.673,-1.352,1.548)	(0.990,-1.716,1.653)	(1.136,-1.881,1.435)
cash	(-0.143,-1.303,0.909)	(0.376,-1.592,1.635)	(2.619,-1.941,2.069)
cashdebt	(0.233,-1.455,1.292)	(1.632,-1.747,1.658)	(-0.097,-1.781,1.749)
cashpr	(-0.598,-1.417,1.159)	(-0.880,-1.597,1.740)	(0.639,-1.732,1.629)
cdi	(0.375,-1.445,1.182)	(2.358,-1.640,1.726)	(1.314,-1.880,1.684)
cei	(-0.403,-1.432,1.751)	(0.644,-1.605,1.755)	(0.224,-1.711,1.926)
cfp	(-0.774,-1.377,1.036)	(1.273,-1.716,1.951)	(1.734,-2.068,1.683)
cfp_ia	(-0.953,-1.447,1.313)	(-0.245,-1.654,1.748)	(1.015,-1.760,1.673)
chatoia	(0.067,-1.462,1.387)	(1.212,-1.680,1.925)	(0.365,-1.946,1.775)
chcsho	(-0.567,-1.374,1.206)	(1.130,-1.472,1.453)	(2.857,-1.723,1.878)
chempia	(-0.113,-1.499,1.120)	(0.927,-1.528,1.752)	(0.967,-1.843,1.978)
chinv	(-0.032,-1.434,0.949)	(0.360,-1.615,1.700)	(1.068,-1.785,2.108)
chmom	(-0.538,-1.493,1.230)	(1.670,-1.647,1.743)	(0.916,-1.814,2.102)
chpmia	(4.210,-1.468,1.144)	(1.422,-1.738,1.931)	(0.614,-1.853,1.624)
chtx	(1.741,-1.467,1.292)	(-0.072,-1.823,1.612)	(0.906,-1.908,1.710)
cinvest	(-0.713,-1.381,1.284)	(0.057,-1.743,1.936)	(-0.576,-2.055,1.734)

cinvest_a	(0.423,-1.360,1.309)	(0.732,-1.687,1.503)	(1.222,-2.057,1.535)
convind	(-0.218,-1.373,1.201)	(0.379,-1.643,1.473)	(1.394,-1.965,2.196)
cp	(-1.133,-1.326,0.972)	(0.468,-1.798,1.718)	(0.490,-1.929,1.694)
cto	(-0.693,-1.467,1.048)	(1.386,-1.721,1.611)	(0.528,-1.815,1.962)
currat	(-0.881,-1.359,1.024)	(1.019,-1.487,1.552)	(2.060,-1.742,1.846)
dcoa	(-0.863,-1.450,1.182)	(-0.243,-1.718,1.655)	(0.135,-1.789,1.784)
dcol	(-0.454,-1.387,0.950)	(1.528,-1.517,1.534)	(1.689,-2.040,1.581)
depr	(-0.659,-1.338,1.138)	(0.288,-1.559,1.572)	(2.824,-1.866,2.116)
dfin	(-0.149,-1.268,1.575)	(1.058,-1.779,1.311)	(1.141,-1.949,1.634)
dfnl	(1.797,-1.460,1.570)	(0.484,-1.841,1.907)	(0.564,-2.142,1.931)
dnca	(3.985,-1.330,1.159)	(1.264,-1.678,1.752)	(0.588,-1.710,1.663)
dncl	(-0.603,-1.386,1.525)	(1.438,-1.661,1.605)	(1.150,-2.074,1.576)
dnco	(-1.272,-1.373,1.426)	(0.749,-1.564,1.382)	(1.294,-1.816,1.684)
dnoa	(1.930,-1.478,1.434)	(0.727,-1.657,1.607)	(0.646,-1.900,1.606)
dolvol	(3.982,-1.396,1.164)	(3.947,-1.795,2.157)	(3.187,-1.739,1.937)
dpia	(1.287,-1.367,1.425)	(0.377,-1.524,1.841)	(1.372,-1.829,1.683)
dsti	(6.661,-1.385,1.647)	(1.338,-1.705,1.759)	(2.237,-1.908,2.245)
dwc	(-0.919,-1.443,1.430)	(0.044,-1.610,1.838)	(1.246,-1.831,1.957)
dy	(-0.831,-1.303,1.263)	(0.802,-1.647,1.777)	(2.258,-1.942,1.637)
ear	(-0.148,-1.448,1.431)	(2.553,-1.788,2.261)	(1.711,-2.020,2.102)
ebp	(0.439,-1.279,1.365)	(-0.127,-1.529,1.339)	(0.869,-1.866,1.698)
egr	(0.078,-1.424,1.134)	(0.399,-1.703,1.278)	(-0.205,-1.918,1.899)
egr_hxz	(-0.415,-1.345,0.931)	(-0.274,-1.660,1.263)	(0.424,-1.853,1.686)
em	(2.163,-1.469,1.016)	(0.920,-1.537,1.297)	(-0.034,-1.757,1.971)
ep	(-0.551,-1.281,1.117)	(0.172,-1.743,1.719)	(3.853,-1.940,1.830)
etr	(1.185,-1.404,1.074)	(0.285,-1.856,1.803)	(0.392,-2.021,1.665)
gad	(2.053,-1.260,1.145)	(0.524,-1.488,1.655)	(0.532,-1.935,1.922)
gma	(0.208,-1.427,1.379)	(0.424,-1.821,1.622)	(0.439,-1.831,1.936)
grcapx	(0.825,-1.459,1.184)	(1.793,-1.749,1.929)	(1.477,-1.874,1.572)
grltnoa	(1.713,-1.427,1.584)	(0.890,-1.468,1.381)	(1.514,-1.865,2.005)
grltnoa_hxz	(2.589,-1.449,1.416)	(1.003,-1.791,1.873)	(0.426,-1.874,1.835)
herf	(1.008,-1.431,0.834)	(3.403,-1.493,1.410)	(2.541,-1.831,1.421)
hire	(-0.735,-1.412,1.041)	(1.083,-1.775,1.208)	(0.485,-1.851,1.869)
idiovol	(1.138,-1.429,1.936)	(1.907,-1.763,1.583)	(2.695,-1.827,1.842)
ill	(1.892,-1.477,1.946)	(0.473,-1.596,1.893)	(0.130,-1.871,1.705)
indmom	(-0.259,-1.378,1.734)	(0.913,-1.528,1.367)	(-0.125,-1.903,1.320)
invest	(1.627,-1.353,1.298)	(0.374,-1.528,1.845)	(1.400,-1.812,1.772)
ivg	(2.508,-1.454,1.167)	(1.442,-1.837,1.640)	(0.760,-1.811,1.978)

kz	(0.467,-1.354,1.887)	(-0.350,-1.587,1.413)	(0.470,-1.724,1.447)
lev	(0.019,-1.366,1.461)	(-0.224,-1.628,1.658)	(2.379,-2.007,1.289)
lfe	(-0.053,-1.397,0.777)	(1.038,-1.639,1.780)	(-0.051,-1.795,1.780)
lgr	(-0.713,-1.425,1.269)	(1.773,-1.653,1.504)	(0.342,-1.869,2.183)
maxret	(0.093,-1.372,1.266)	(0.846,-1.890,1.537)	(2.429,-1.806,1.711)
mom36m	(1.847,-1.475,1.349)	(1.609,-1.816,1.573)	(0.730,-1.775,2.041)
mom6m	(0.602,-1.499,1.556)	(0.998,-1.579,1.689)	(0.315,-1.944,1.698)
moms12m	(2.055,-1.422,1.301)	(0.226,-1.669,1.406)	(0.512,-1.736,1.689)
ms	(-1.084,-1.470,1.506)	(0.621,-1.714,1.997)	(0.607,-1.803,1.864)
ndf	(2.431,-1.455,1.218)	(1.540,-1.934,1.528)	(0.188,-1.759,1.732)
ndp	(-0.875,-1.273,1.070)	(0.658,-1.649,1.560)	(1.864,-1.997,1.433)
nef	(-0.484,-1.425,1.854)	(2.485,-1.701,1.819)	(1.688,-1.779,1.794)
nincr	(-0.504,-1.328,0.950)	(0.656,-1.732,1.789)	(0.903,-1.862,1.541)
noa	(0.553,-1.334,1.533)	(0.485,-1.725,1.517)	(3.193,-1.984,2.069)
nop	(-0.455,-1.387,1.242)	(-0.188,-1.716,1.624)	(1.561,-1.654,1.760)
nxf	(-0.016,-1.428,1.352)	(0.991,-1.843,1.629)	(1.349,-1.717,1.691)
ob_a	(1.603,-1.495,1.444)	(0.231,-1.594,2.015)	(0.091,-1.803,1.660)
ol	(0.086,-1.449,1.515)	(1.779,-1.540,1.677)	(0.114,-1.723,1.867)
op	(-0.152,-1.328,1.382)	(0.315,-1.713,1.726)	(1.493,-1.673,1.661)
orgcap	(1.450,-1.435,1.491)	(0.656,-1.617,1.798)	(0.745,-1.663,1.797)
os	(-0.989,-1.526,1.655)	(1.369,-1.816,1.653)	(-0.872,-1.881,1.594)
pchcapx	(-0.450,-1.445,1.065)	(0.991,-1.691,1.507)	(0.607,-1.728,1.592)
pchcapx3	(-0.378,-1.426,1.274)	(1.269,-1.624,1.586)	(0.331,-1.799,1.575)
pchcapx_ia	(1.596,-1.448,1.496)	(2.784,-1.684,1.720)	(-0.103,-1.843,1.603)
pchcurrat	(4.353,-1.367,2.208)	(2.972,-1.669,2.077)	(2.672,-1.920,1.522)
pchdepr	(-0.793,-1.424,1.345)	(0.803,-1.607,1.474)	(2.019,-1.939,1.753)
pchgm_pchsale	(3.384,-1.384,1.866)	(1.614,-1.671,1.804)	(1.302,-1.677,2.320)
pchquick	(4.237,-1.468,1.700)	(2.500,-1.912,1.986)	(1.445,-1.935,2.296)
pchsale_pchinvt	(0.521,-1.531,1.597)	(1.397,-1.807,1.345)	(0.120,-1.969,1.465)
pchsale_pchrect	(1.579,-1.496,1.318)	(1.750,-1.882,1.303)	(2.090,-1.944,1.711)
pchsale_pchxsga	(0.891,-1.423,1.663)	(0.213,-1.817,1.429)	(0.479,-1.883,1.906)
pchsaleinv	(-0.371,-1.520,1.165)	(1.474,-1.794,1.679)	(-0.062,-1.999,1.669)
pctacc	(0.130,-1.506,1.387)	(-1.155,-1.924,1.603)	(1.266,-2.025,2.072)
pm	(-0.332,-1.486,1.877)	(0.425,-1.702,1.388)	(-0.126,-1.698,1.611)
poa	(0.028,-1.392,1.418)	(-0.399,-1.887,1.653)	(2.095,-1.673,2.133)
pps	(-0.554,-1.467,1.521)	(2.088,-1.699,1.493)	(1.316,-2.000,1.656)
pricedelay	(0.048,-1.438,1.155)	(1.231,-1.630,1.491)	(1.270,-1.676,1.757)
ps	(0.070,-1.402,1.013)	(1.720,-1.626,1.639)	(1.852,-1.923,2.399)
quick	(-0.454,-1.298,1.320)	(0.104,-1.660,1.626)	(2.042,-1.855,1.646)
rd	(1.235,-1.385,1.293)	(0.686,-1.605,1.393)	(2.045,-2.052,1.939)

rdm	(-0.668,-1.334,1.046)	(0.997,-1.653,1.659)	(1.644,-1.846,2.310)
rds	(-0.294,-1.336,1.278)	(1.032,-1.641,1.867)	(1.422,-1.871,2.003)
realestate_hxz	(4.818,-1.407,1.671)	(1.314,-1.704,2.302)	(1.895,-1.874,1.946)
retvol	(0.538,-1.374,1.410)	(1.947,-1.664,1.994)	(2.378,-1.646,1.813)
rna	(0.345,-1.416,1.486)	(2.723,-1.552,1.425)	(1.579,-1.873,1.738)
roaq	(-0.235,-1.423,1.477)	(0.760,-1.773,1.949)	(1.355,-1.780,1.933)
roavol	(-0.549,-1.298,1.444)	(0.403,-1.642,1.332)	(2.956,-1.891,1.838)
roic	(-0.346,-1.526,1.441)	(0.438,-1.828,1.678)	(0.780,-1.938,1.941)
rs	(0.145,-1.444,1.043)	(0.894,-1.679,1.380)	(1.696,-1.760,1.973)
rsup	(7.620,-1.660,1.791)	(3.774,-1.879,1.396)	(4.245,-1.892,1.760)
salecash	(-0.498,-1.295,0.905)	(0.618,-1.535,1.481)	(3.306,-1.794,2.261)
saleinv	(0.494,-1.353,1.405)	(2.000,-1.548,1.529)	(1.280,-1.839,2.006)
salerec	(1.576,-1.294,1.621)	(1.666,-1.604,1.792)	(0.488,-1.813,1.652)
sgr	(-0.569,-1.402,1.068)	(1.314,-1.628,1.923)	(-0.166,-1.852,1.961)
sin	(0.813,-1.396,1.686)	(1.965,-1.765,1.813)	(0.323,-1.829,1.820)
sp	(6.008,-1.436,2.234)	(0.889,-1.684,1.648)	(1.824,-1.710,1.576)
std_dolvol	(-0.574,-1.414,1.171)	(1.399,-1.740,2.066)	(0.797,-1.710,2.049)
std_turn	(-0.369,-1.318,1.343)	(0.613,-1.854,1.429)	(2.124,-1.827,1.973)
stdacc	(-0.192,-1.387,1.411)	(0.677,-1.696,1.703)	(1.687,-1.969,1.753)
stdcf	(0.166,-1.382,1.550)	(0.742,-1.651,1.527)	(1.588,-1.948,1.875)
sue	(0.730,-1.418,1.830)	(-1.173,-1.767,1.553)	(-0.879,-1.910,2.054)
ta	(1.982,-1.441,1.655)	(0.716,-1.696,1.552)	(0.912,-1.926,1.734)
tang	(0.015,-1.353,1.349)	(1.272,-1.619,1.896)	(2.629,-1.869,1.523)
tb	(-0.629,-1.390,1.370)	(2.196,-1.545,1.728)	(2.121,-1.693,1.844)
turn	(0.156,-1.331,1.214)	(0.650,-1.736,1.453)	(1.835,-1.791,1.782)
ww	(-0.587,-1.381,1.577)	(1.681,-1.692,1.802)	(2.706,-2.022,1.559)
zerotrade	(0.231,-1.380,1.258)	(0.843,-1.765,1.632)	(2.054,-1.786,1.808)
zs	(0.212,-1.323,1.744)	(-0.160,-1.568,1.545)	(2.027,-2.099,1.271)

2 Log

2.1 2020-09-01

$$\hat{u}_i = \bar{Y}_i - \hat{b}_i'\hat{\gamma} = \bar{\varepsilon}_i - \left(\hat{b}_i - b_i\right)'\bar{\gamma} - \hat{b}_i'(\hat{\gamma} - \bar{\gamma}) \tag{9}$$

Roughly $\bar{\varepsilon}_i$ will be kept, the last term $\bar{\gamma} - \hat{\gamma}$ is of smaller order, the second term will contribute.

Let $R_2 = \left(\hat{b}_i - b_i\right)'\bar{\gamma}$,

Lemma 2.1 (R_2)**.** $R_2 = \dots$

Proof.

$$R_2 = \left(\Sigma_F^{-1} + o_p(1)\right)\frac{1}{T}\sum_t \varepsilon_{it}\left(F_t - \bar{F}\right) \tag{10}$$

$$= \frac{1}{T}\sum_t \varepsilon_{it}\Sigma_F^{-1}\left(F_t - \bar{F}\right)\bar{\gamma} \tag{11}$$

$\Sigma_F^{-1}\left(F_t - \bar{F}\right)\bar{\gamma} = \Sigma_F^{-1}\left(F_t - \bar{F}\right)\left(\gamma + \bar{F} - EF\right)$, let $c_t = \Sigma_F^{-1}\left(F_t - \bar{F}\right)\gamma$. We will have

$$\hat{u}_i = \frac{1}{T}\sum_t (1 - c_t)\varepsilon_{it} + R_3 + R_4 \tag{12}$$

where $R_3 = \hat{b}_i'(\bar{\gamma} - \hat{\gamma})$, R_4 comes from the estimate of c_t.

c_t is independent of ε_{it} for all i, t. □

We construct based on the following statistic:

$$\mathbf{T}_{n,T} = \frac{1}{n(n-1)}\sum_i \sum_{j \neq i} \hat{u}_i \hat{u}_j \frac{1}{h} K\left(\frac{\hat{b}_i - \hat{b}_j}{h}\right) \tag{13}$$

we show that under scaling it converges in distribution.

The draft is

$$\mathbf{T}_{n,T} = \frac{1}{n(n-1)T^2}\sum_t (1-c_t)\sum_{i\neq j}\left[\varepsilon_{it}\varepsilon_{jt}\frac{1}{h}K\Big(\hat{b}_i - \hat{b}_j/h\Big)\right] \tag{14}$$

$$= \frac{1}{n(n-1)T^2}\sum_t (1-c_t)\sum_{i\neq j}\left[\varepsilon_{it}\varepsilon_{jt}\frac{1}{h}\big(K\big(b_i - b_j/h\big)\big) + R_K\right] \tag{15}$$

$$= \frac{1}{\sqrt{n(n-1)}T^2}\sum_t (1-c_t)\frac{1}{\sqrt{n(n-1)}h}\sum_{i\neq j}\varepsilon_{it}\varepsilon_{jt}K\left(\frac{b_i - b_j}{h}\right) \tag{16}$$

My guess is that

$$nT^{\frac{3}{2}}\mathbf{T} \rightsquigarrow N(\mu, \text{var}) \tag{17}$$

3 Inference

Let's first assume there is no s_i. We write vectors in columns.

We construct the test statistic using the two-pass regression method. We calculate the residual estimated under the null specification as well as the nonparametric alternative and compare the difference.

There are two procedures commonly used:

1. the cross-sectional regression approach: where we first estimate the expected excess return with $\bar{Y}_i = \frac{1}{T}\sum_{t=1}^{T} Y_{it}$ and $\bar{F} = \frac{1}{T}\sum_{t=1}^{T} F_t$ and then run a single cross-sectional regression. Shanken, 1992.

2. the Fama-MacBeth runs a cross-sectional regression for each time t, obtain the regression estimate $\hat{\gamma}_t$ for each t and the use $\hat{\gamma}_t$ as a sample for γ. This corrects for the cross-sectional correlation but not the error from first step estimation of β.

3.1 Estimation Residual Under the Null

First we estimate each stock's β's via the time-series regression:

$$Y_i = F\beta_i + \varepsilon_i \quad \text{for} \quad i = 1, \ldots, N$$

where F is $T \times k$ and $Y_i : T \times 1$.

$$\hat{\beta}_i = \beta_i + \left(F'F\right)^{-1} F'\varepsilon_i$$

In the cross-sectional regression, under the null we will have the following regression model for each time t,

$$Y_{it} = \beta_i'\gamma + u_{it} \tag{18}$$

with

$$u_{it} = Y_{it} - \beta_i'\gamma_t = \varepsilon_{it} + \beta_i'(F_t - E(F))$$

which satisfies

$$E(u_{it} \mid \beta_i) = 0.$$

Let $\beta = (\beta_1, \ldots, \beta_N)'$ which is $N \times k$. Equation 18 can be written in the matrix form as

$$Y_t = \beta\gamma_t + u_t \tag{19}$$

Suppose we replace β with $\hat{\beta}$ from first step and estimate the γ using cross-sectional OLS then

$$\hat{\gamma}_t = \left(\hat{\beta}'\hat{\beta}\right)^{-1}\hat{\beta}'Y_t = \left(\hat{\beta}'\hat{\beta}\right)^{-1}\hat{\beta}'[\beta F_t + \varepsilon_t]$$

Then

$$\begin{aligned}\hat{u}_{it} &= Y_{it} - \hat{\beta}_i'\hat{\gamma} \\ &= \beta_i' F_t + \varepsilon_{it} - \hat{\beta}_i'\hat{\gamma}_t\end{aligned}$$

Stack the $\hat{u}_{it}$ into $\hat{u}_t$ we have

$$\begin{aligned}\hat{u}_t &= \beta F_t + \varepsilon_t - \hat{\beta}\left(\hat{\beta}'\hat{\beta}\right)^{-1}\hat{\beta}'[\beta F_t + \varepsilon_t] \\ &= \left(I - \hat{\beta}\left(\hat{\beta}'\hat{\beta}\right)^{-1}\hat{\beta}'\right)[\beta F_t + \varepsilon_t] \\ &= \left(I - \hat{\beta}\left(\hat{\beta}'\hat{\beta}\right)^{-1}\hat{\beta}'\right)\left[u_t - \left(\hat{\beta} - \beta\right)\gamma_t\right]\end{aligned}$$

Let $\bar{\varepsilon}_i = \frac{1}{T}\sum_t \varepsilon_{it}$ or ε_{it} corresponding to cross-sectional regression or Fama-

MacBeth. $\bar{\gamma} = \gamma + \bar{F} - E(F)$, we have

$$\begin{aligned}\hat{u}_i &= \bar{\varepsilon}_i - \hat{\beta}_i \hat{A}_\beta \bar{\varepsilon} - \left\{\left(\hat{\beta}_i - \beta_i\right)\bar{\gamma} - \hat{\beta}_i \hat{A}_\beta \left(\hat{\beta} - \beta\right)\bar{\gamma}\right\} \\ &= \begin{cases} O_p(1) + O_p\left(\frac{1}{\sqrt{N}}\right) + O_p\left(\frac{1}{\sqrt{T}}\right) + O_p\left(\frac{1}{\sqrt{NT}}\right) & \text{if we don't do the time averaging first} \\ O_p(\frac{1}{\sqrt{T}}) + O_p\left(\frac{1}{\sqrt{NT}}\right) + O_p\left(\frac{1}{T}\right) + O_p\left(\frac{1}{\sqrt{NT}}\right) & \text{if we do the time averaging first} \end{cases}\end{aligned}$$

3.2 Estimation Residual Under the Alternative

Assume we have

$$m(x) := E(Y_i \mid \beta_i = x)$$

Then by similar procedure we have

$$\hat{m}(x) = \left(\hat{X}_D' \hat{W}_D \hat{X}_D\right)^{-1} \hat{X}_D' \hat{W}_D \bar{Y} = \left(\hat{X}_D' \hat{W}_D \hat{X}_D\right)^{-1} \hat{X}_D' \hat{W}_D \left[\bar{M} + \bar{\varepsilon}\right]$$

where $\hat{X}$ is the estimated betas and $\hat{W}$ is a diagonal matrix consisting of the K_h.

We embed the model in a nonparametric regression model and estimate with local linear method.

Follow the procedure in Li and Racine, 2007 or Hardle and Mammen, 1993, we test the specification based on the error term $\hat{u}_t = Y_t - \hat{\beta}\hat{\gamma}$, under H_0 we have:

$$E(uE(u \mid \beta = b) f(b)) = 0 \quad a.s.$$

Following Li and Racine, 2007, we construct a Nadaraya-Watson type kernel regression estimator. Let $K_{h,ij} = \prod_{i=1}^{k} k\left(\left(X_i - X_j\right)/h\right)$, then we try to use the relationship:

$$E[uE(u \mid \beta = b) f(b)] = 0 \quad a.s. - b \quad \text{under} \quad H_0$$

Our test statistic is based on a transformed and studentized version of the following statistic:

$$I = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j \neq i} \hat{u}_i \hat{u}_j K_{h,ij}$$

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